



POLITECNICO
MILANO 1863 | DIPARTIMENTO
DI MECCANICA

Project presentation

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Goal

Teach long manipulation sequences to a robotic arm leveraging **Vision-Language Models (VLMs)**

Inputs:

- Images
- Textual instructions

You are asked to:

1. **Teach** a series of **pick and place** actions to the robot
2. **Validate** the task plan
3. **Test** the robot policy **without human intervention**

You will perform simulations and real-world experiments

Goal

Intended learning objectives

By the end of this lab, you will be able to:

1. Understand what a Vision-Language Model is and how it works.
2. Use an API-based model from Hugging Face.
3. Process visual input (RGB image) and textual input.
4. Convert high-level instructions (e.g., “pick the red cube”) into robot actions.
5. Execute a task pipeline in simulation.
6. Deploy the same logic to a real ABB GoFa robot.
7. Evaluate limitations and failure cases.

Applications:

- Pick and place
- Sorting objects based on color or shape
- Pick up the requested object

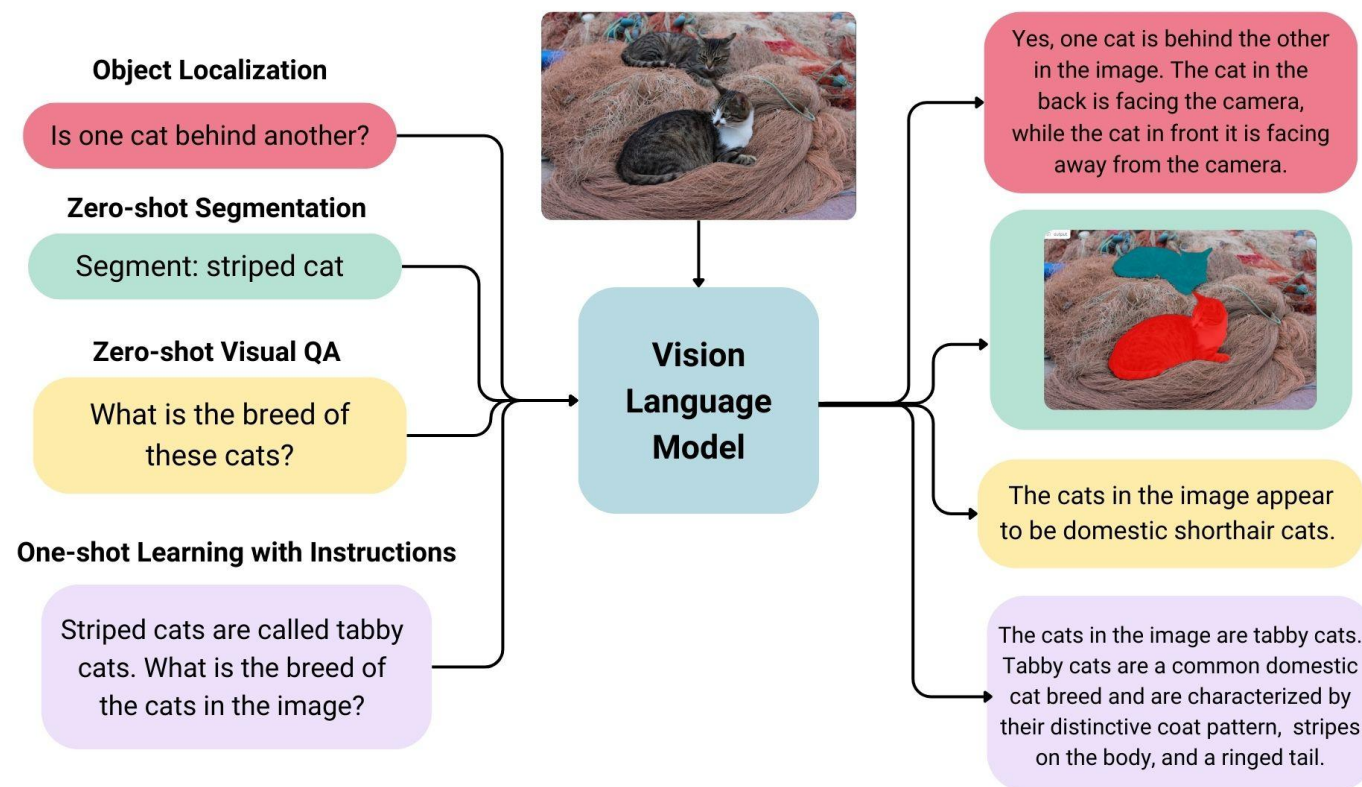
Vision-Language Models

VLMs are a type of **generative models** that take image and text inputs and generate text outputs.

They can be used for many different tasks:

- Image recognition
- Visual question answering
- Document understanding
- Bounding boxes
- Segmentation masks

VLMs can also be used to tell a robot **what object** to pick and **where it must be placed**.



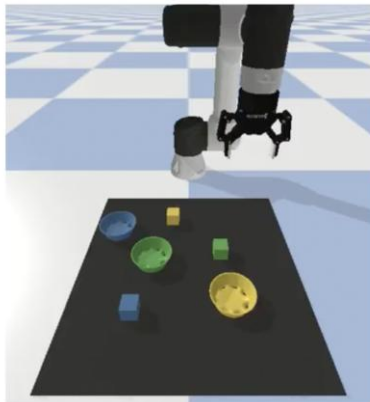
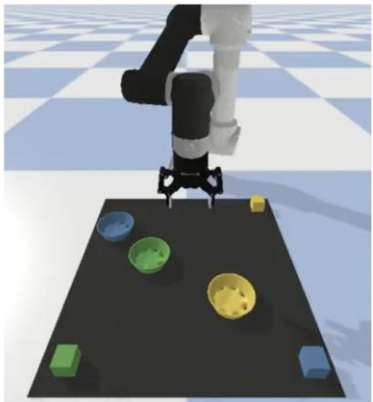
Generative models and task planning

Decompose high-level, abstract goals (e.g., "clean the room") into a **logical, sequential** set of steps (e.g., "navigate to X" "pick up X" "deposit X into Y").

It tells the robot:

- Where to move (x, y, z)
- In an ordered and logical sequence

Recent applications try to use **VLMs for task planning**



Task: Move all the blocks to different corners.

SayCan:

1. Pick up the green block and place it in the lower left corner.
2. Pick up the blue block and place it in the lower right corner.
3. Pick up the yellow block and place it in the upper right corner.

<https://say-can.github.io/>



<https://www.youtube.com/watch?v=TGNFDeohW6o&t=22s>

Simulation environment

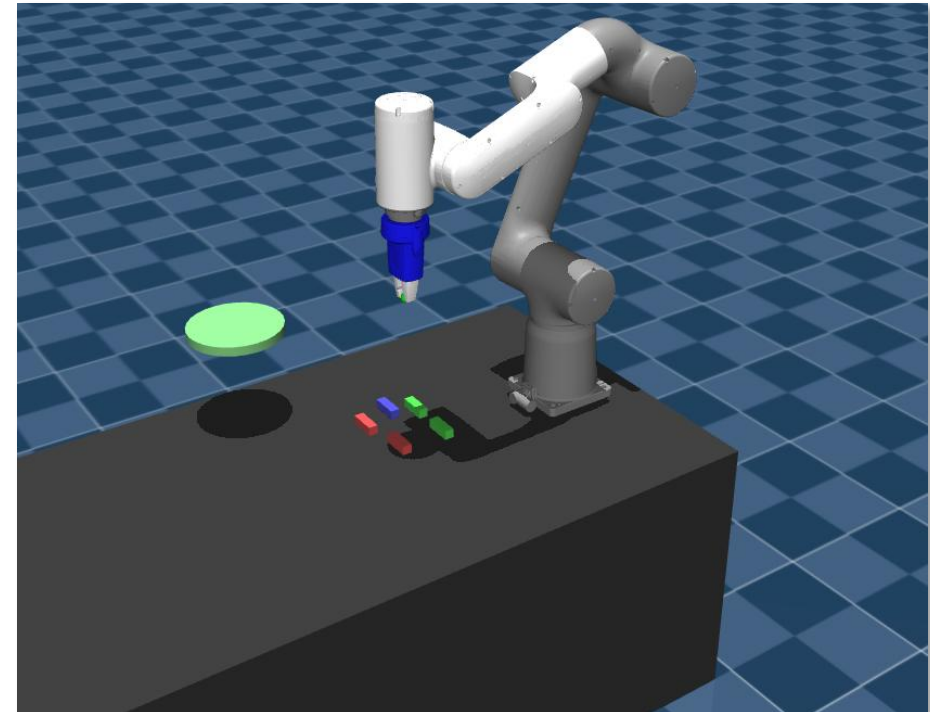
- MuJoCo environment with the **ABB GoFa robot arm** and various objects on the table.
- It is fundamental to test the robot policy before deploying it to the real world

How to use it?

- See tutorial on the provided Colab notebook
- Check MuJoCo documentation

<https://mujoco.org/>

https://github.com/artuurog/GoFa_MuJoCo_sim/tree/main

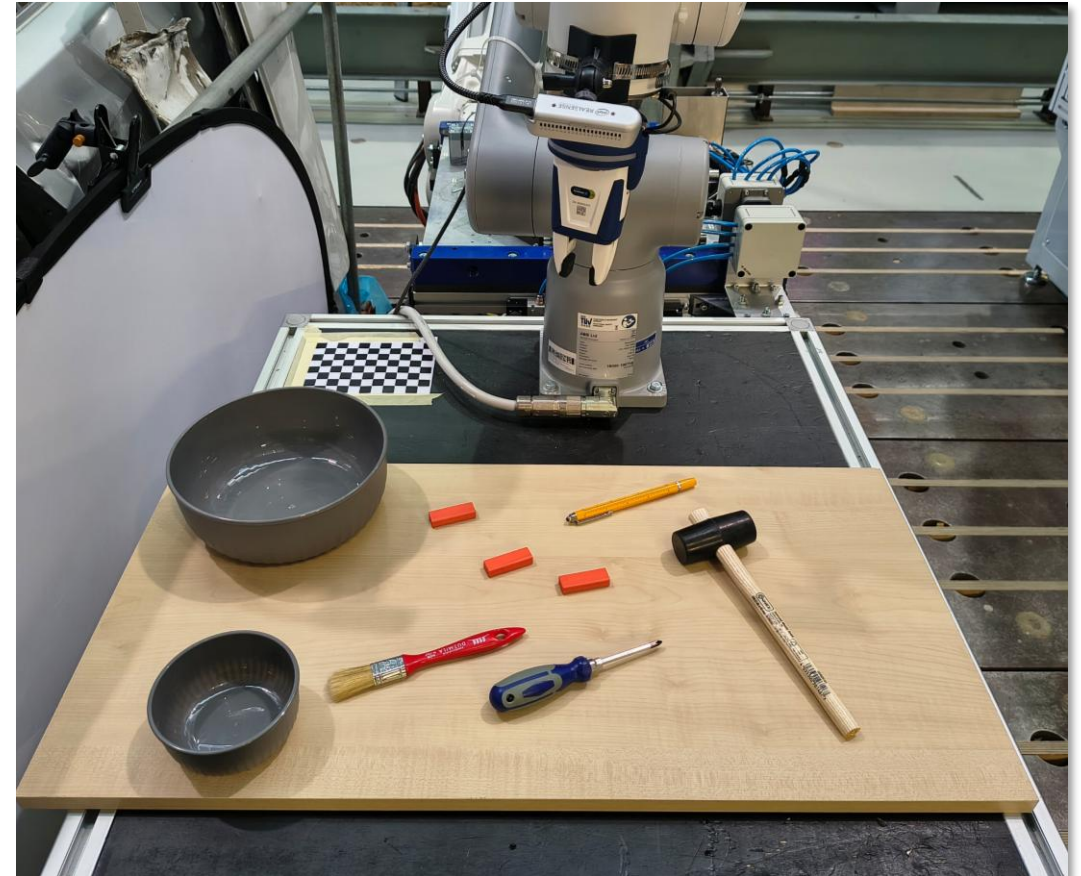
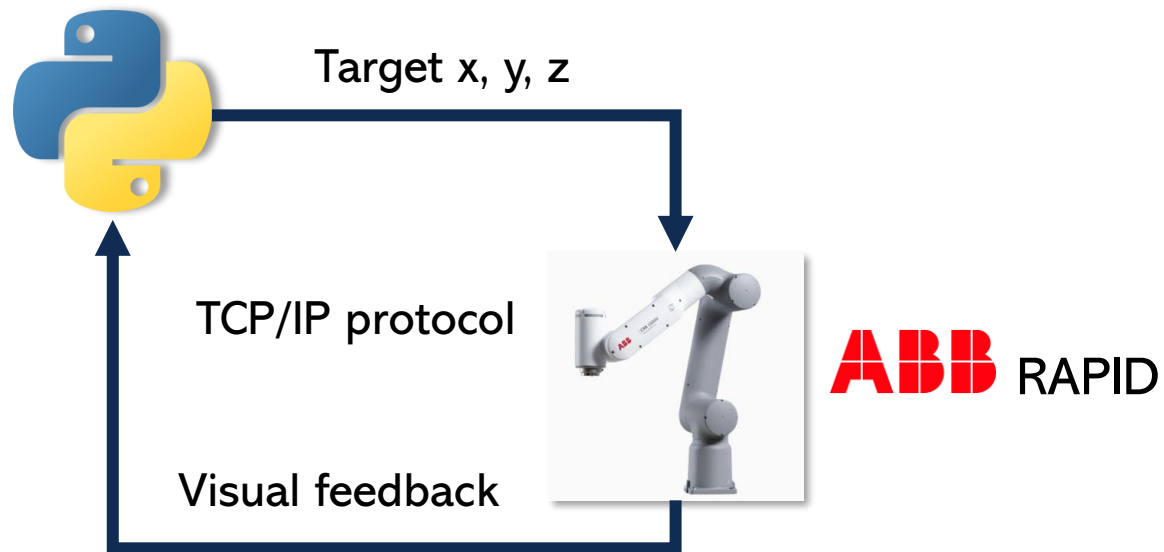


Real robot

Once you are satisfied with the simulations, you can proceed with real world experiments

The robot is an ABB GoFa manipulator arm

Low level control:



Workflow - Training phase

Inputs

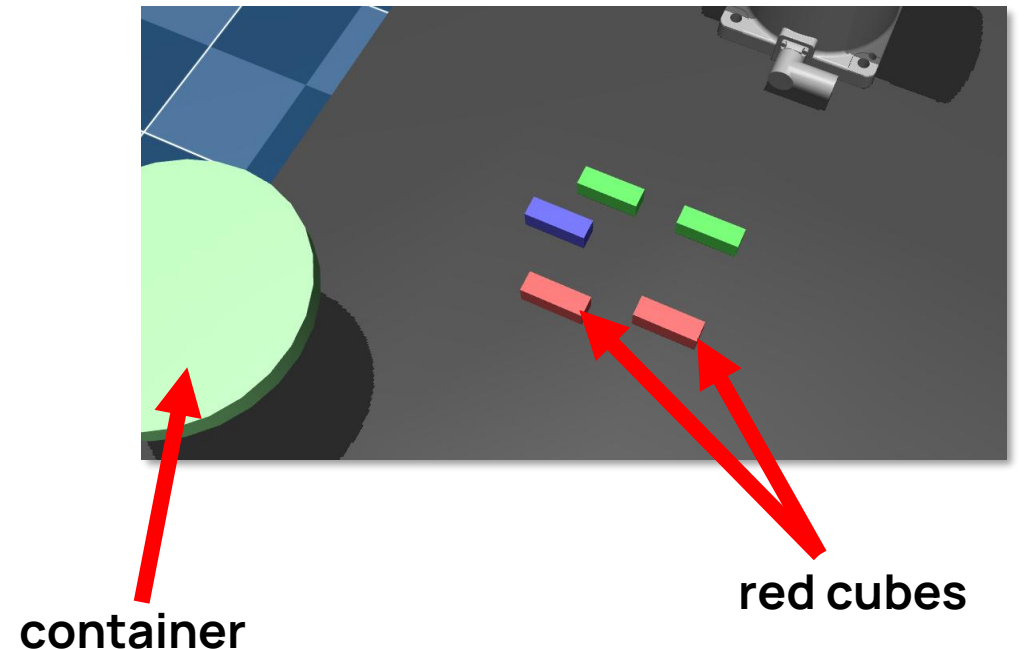
- Human instruction (e.g. «Pick the red cube and place it in the container», «clear the table»)
- Image of the workspace

Let a Vision-Language Model (VLM) reason about:

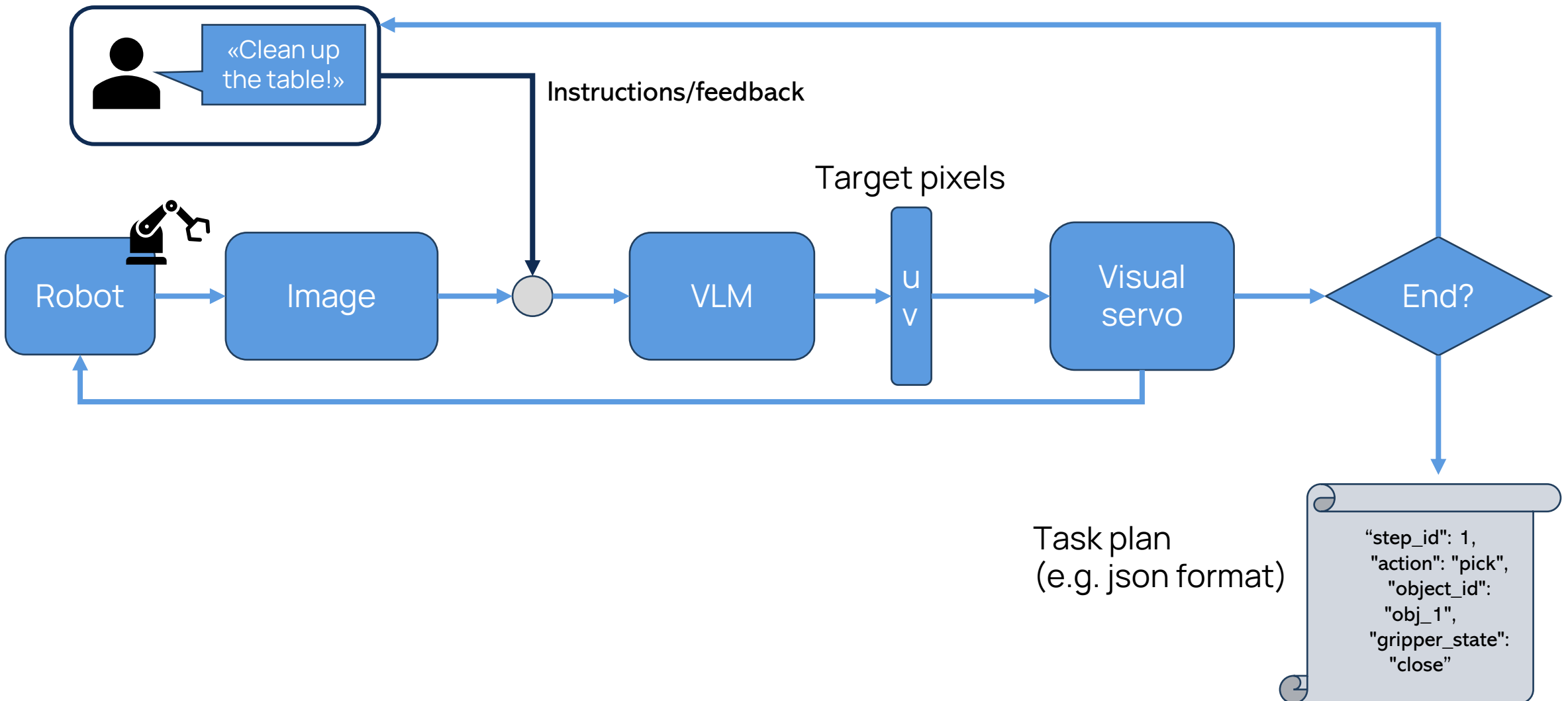
1. Decompose task into subtasks
2. Identify the target to reach
3. Save the operation to a task plan file (e.g. json file)

The human:

- Gives textual instructions to the robot
- Provides corrective feedback if needed



Training phase



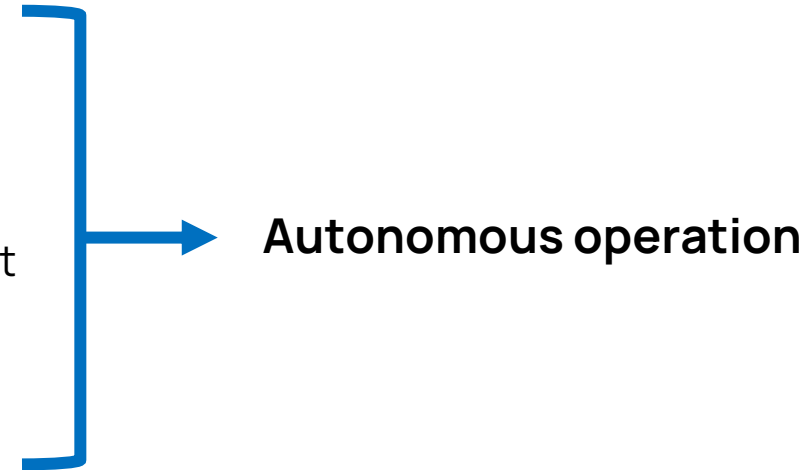
Test phase

Inputs:

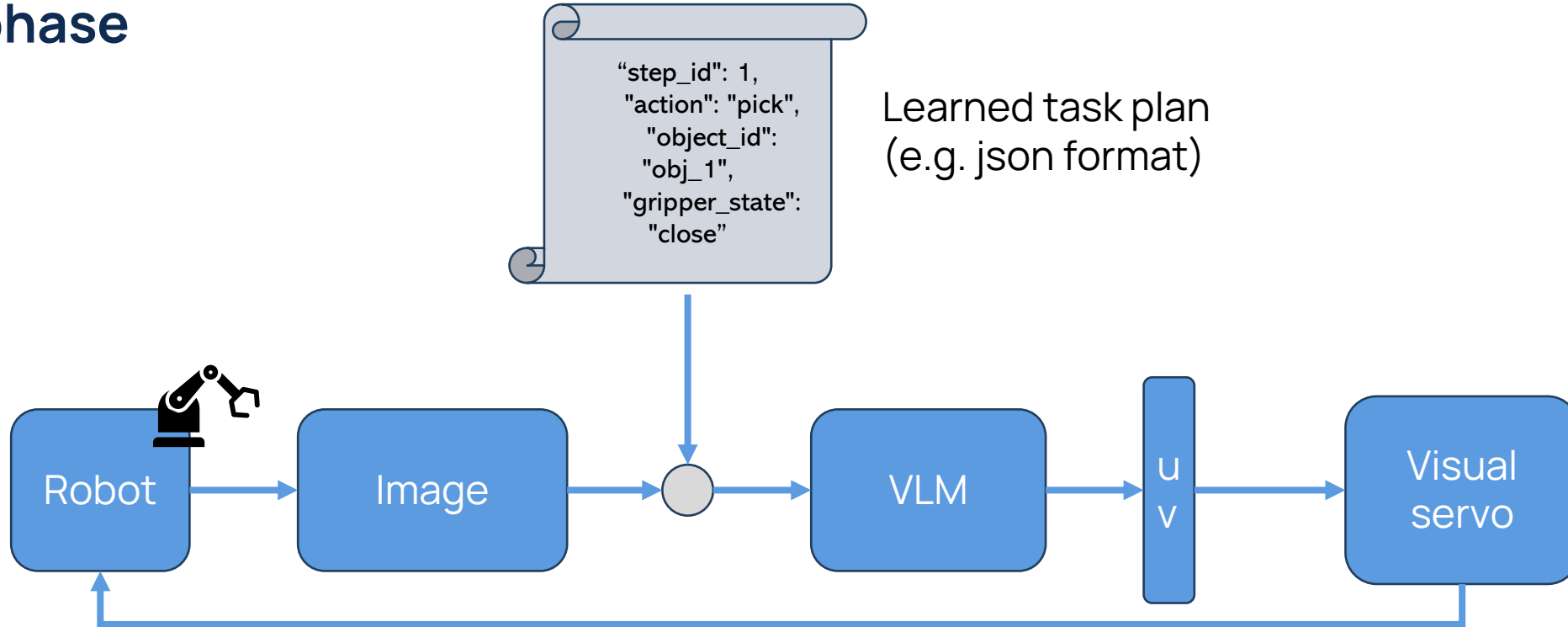
- Task plan
- Image of the workspace

The robot:

- Takes a picture of the objects on the workspace
- Gives the learned task plan + image to the VLM
- Uses a VLM to infer the object to pick and where to place it



Test phase



The robot controller has to:

- Understand the step of the operation based on images, robot states...
- Localize the target to reach
- Open or close the gripper?
- Has the robot reached the target?

Robustness analysis

Object detection

- Is the VLM able to correctly locate the objects in the image? Is an additional object detection module needed?

Spatial precision

- Is the VLM precise enough to guide the robot? Do you need scaffolding or visual prompting for better guidance?

Reasoning ability of the models

- Is the VLM able to elaborate and decompose complex instructions? Can it elaborate implicit requests (e.g. «I'm thirsty», «Mount the wheel on the shaft»)?

Sim-to-real

- If I learn one plan in simulation, can I use it as it is in a real-world scenario?

Robustness analysis

Models

- Compare the performance of different models and sizes.
- Which one can better interpret implicit requests?
- Which one generates the most detailed robot instructions?

Robust control loop

- What happens if you move objects suddenly during the test?
- Does the robot update its target?
- Is it able to adapt in real time to unexpected situations?

References and Materials

Visual servo: <https://faculty.cc.gatech.edu/~seth/ResPages/pdfs/HutHagCor96.pdf>



documentation: <https://mujoco.readthedocs.io/en/stable/overview.html>



HuggingFace: <https://huggingface.co/>



Github repo: https://github.com/artuurog/Machine_Learning_LAB

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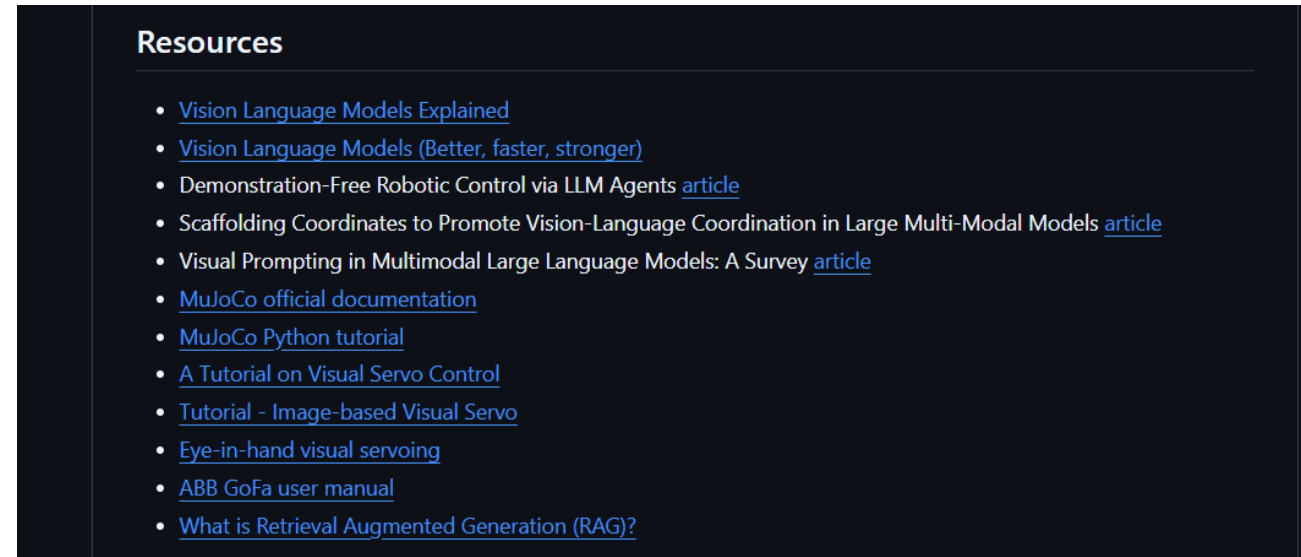
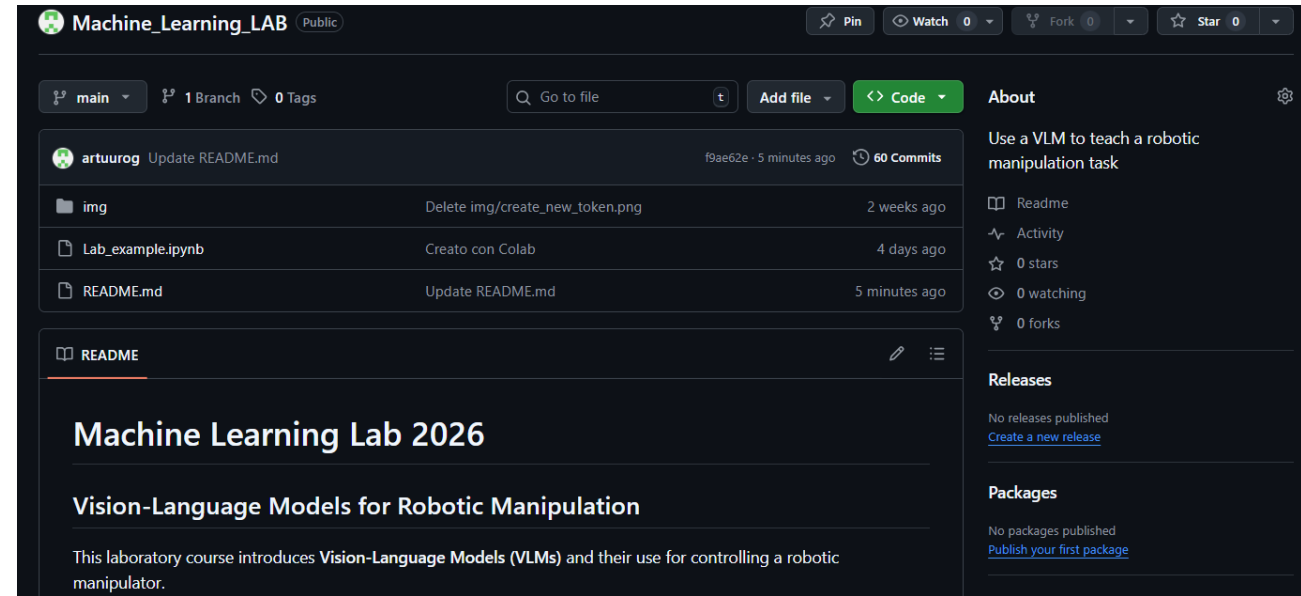
Github repo

https://github.com/artuurog/Machine_Learning_LAB

Here you can find a more detailed description of the project

The repository contains:

- Colab notebook (.ipynb file) with a brief tutorial
- Guides and manuals
- External resources
- Useful links
- Articles to start your research





Thank you!